Smart Home AIoT – Test Data Collection Sheet – Phase F – GPS Welcome/Goodbye

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Instructions for use:
1. Run each test 10 times consecutively (or across sessions, as practical)
2. Record: Date, Time, exact Input given, Actual Output observed, Response Time (use stopwatch), and Pass/Fail
3. Pass criteria: Actual matches Expected description
4. After 10 trials, calculate: Pass Rate %, Average Response Time, Standard Deviation
5. Use Comments for any anomalies or special observations

Statistical formulas:
- Pass Rate = (Pass Count / 10) × 100%
- Avg RT = (response times) / 10
- Variance = (xi -)² / n
- Std Dev = √variance

Coverage Matrix

Phase	Description	Count
Phase F	Phase F – GPS Welcome/Goodbye	8
	TOTAL	8

Phase F – GPS Welcome/Goodbye

F01	GPS Welcome – Garage Door Open	Phase F

Method: Mock person home

Expected: cover.garage_door opening

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10	Pass Rate: ____%	Avg RT: ____ ms	Std Dev: ____ ms

Comments:

F02 GPS Welcome – Garage Lights (LDR-aware) Phase F

Method: garden_ldr > 2400 + welcome

Expected: garage_light + pathway ON

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10	Pass Rate: ____%	Avg RT: ____ ms	Std Dev: ____ ms

Comments:

F03 GPS Welcome – Garden Lights Phase F

Method: garden_ldr > 2200 + welcome

Expected: garden_light + pathway ON

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10	Pass Rate: ____%	Avg RT: ____ ms	Std Dev: ____ ms

Comments:

F04	GPS Welcome – Switch Away Home	Phase F

Method: Set Away + GPS welcome

Expected: house_mode Home auto

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10	Pass Rate: ____%	Avg RT: ____ ms	Std Dev: ____ ms

Comments:

F05	GPS Welcome – LINE Notify	Phase F

Method: GPS welcome

Expected: LINE "ยินดีต้อนรับ"

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10	Pass Rate: ____%	Avg RT: ____ ms	Std Dev: ____ ms

Comments:

F06	GPS Goodbye – 2-min grace	Phase F

Method: person not_home + wait 2 min

Expected: house_mode Away

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10	Pass Rate: ____%	Avg RT: ____ ms	Std Dev: ____ ms

Comments:

F07	GPS Goodbye – Cascade	Phase F

Method: Goodbye trigger

Expected: switches OFF + lock + curtain close

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10	Pass Rate: ____%	Avg RT: ____ ms	Std Dev: ____ ms

Comments:

F08 GPS Goodbye – LINE Notify Phase F

Method: Goodbye trigger

Expected: LINE goodbye message (random variant)

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10	Pass Rate: ____%	Avg RT: ____ ms	Std Dev: ____ ms

Comments: